Data Structures and Algorithms -3 (25CS2103E)

Student-to-Laboratory Allocation System Using Maximum Bipartite Matching

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Abstract

The **Student-to-Laboratory Allocation System** is an automated application developed to efficiently assign students to laboratories based on their preferences while considering laboratory capacity constraints. Traditional manual allocation methods are often time-consuming, prone to errors, and may not provide fair assignments. To overcome these challenges, this project utilizes the **Maximum Bipartite Matching** algorithm, a graph-based approach that determines the optimal allocation by maximizing the number of successful student-to-laboratory assignments.

In this system, students submit their preferred laboratories, and the administrator manages laboratory details such as capacity and availability. The application models students and laboratories as two disjoint sets of a bipartite graph, where edges represent student preferences. The matching algorithm ensures that each student is assigned to **at most one laboratory** while maximizing the total number of successful allocations without exceeding laboratory capacities.

The system also generates allocation reports, displays allocated and unallocated students, and provides an efficient and transparent scheduling process. By automating laboratory allocation, the project reduces manual effort, minimizes conflicts, and ensures fair resource utilization. This project demonstrates the practical application of graph matching algorithms in solving real-world scheduling and resource allocation problems, making it an effective solution for educational institutions.

Introduction

The **Student-to-Laboratory Allocation System** is a console-based application that automatically assigns students to laboratories based on their preferences and available laboratory capacity. The system uses the **Maximum Bipartite Matching** algorithm to maximize the number of successful student-laboratory assignments.

Students provide their preferred laboratories, while laboratory capacities are provided as input. The system processes these details and displays the allocated and unallocated students along with laboratory-wise allocation results. This project demonstrates the practical application of **graphs, DFS, and bipartite matching** in solving real-world resource allocation problems.

Literature Survey

Existing System	Method Used	Limitation
Manual Allocation	Manual Assignment	Time Consuming
Preference-based Allocation	Student Preferences	May not maximize assignments
Graph-Based Allocation	Bipartite Graph	Requires Algorithmic Processing

Problem Statement and Objectives

Problem Statement:

To develop an efficient system that allocates students to laboratories based on their preferences and laboratory capacity while maximizing the number of successful assignments using Maximum Bipartite Matching.

Objectives:

- Automate laboratory allocation.
- Consider student preferences.
- Handle laboratory capacity constraints.
- Maximize successful allocations.
- Minimize allocation conflicts.
- Apply Maximum Bipartite Matching in a real-world problem.

Flow Chart

Student Registration

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Student Selects Lab Preferences

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Admin Adds Laboratories

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Admin Sets Capacity

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Store Data in Database

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Create Bipartite Graph

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Run Maximum Bipartite Matching

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Allocate Students

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Display Results

↓

Generate Reports

Results

- ➔ Successfully allocated the **maximum possible number of students**.
- ➔ Student preferences were considered during allocation.
- ➔ Laboratory capacity constraints were maintained.
- ➔ Each student was assigned to **at most one laboratory**.
- ➔ Students who could not be assigned were clearly identified
- ➔ Lab-wise allocation results were displayed.
- ➔ The system calculated the overall **allocation percentage**.

Hardware and Software requirements

Software Requirements

- Windows/Linux/macOS
- **Java JDK 17+**
- **Visual Studio Code / IntelliJ IDEA**
- Java Collections Framework
- Console/Terminal

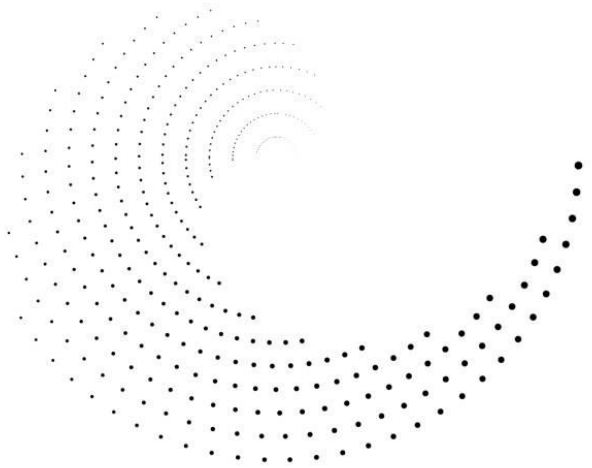
Conclusion

The **Student-to-Laboratory Allocation System** successfully provides an efficient method for assigning students to laboratories based on their preferences and available capacity. By using the **Maximum Bipartite Matching algorithm**, the system maximizes the number of successful allocations while avoiding conflicts. The project demonstrates the practical application of **graphs, DFS, and matching algorithms** in solving a real-world resource allocation problem .

Future Scope

The project can be enhanced in the future by adding:

- **Priority-based allocation** based on CGPA, year, or other criteria.
- **Multiple laboratory sessions** and time-slot management.
- **Admin and student login** functionality.
- **Database integration** for storing student and laboratory data.
- **Waitlist management** for unallocated students.
- **Automatic report generation** for allocation results.
- **Web or mobile interface** for easier access.
- **Reallocation functionality** when students or lab capacities change.



Thank you